THORACIC MOVEMENT

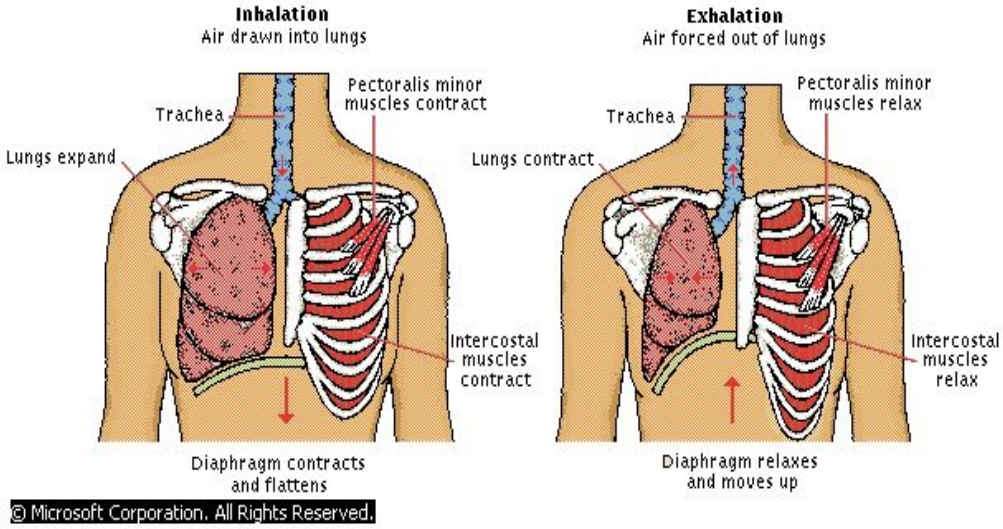
LEARNING OBJECTIVES

At the end of the lecture the student should be able to know:

- About principles of respiratory movement
- Movements involved to change diameter of thoracic cage
- Movement in different phases of respiration, both under normal and stressed condition

PRINCIPLE OF THORACIC MOVEMENTS

- The lungs expand passively during inspiration and retract during expiration
- These movements are governed by the following two factors.
  - Alterations in the capacity of the thorax
  - Elastic recoil of the pulmonary alveoli and of the thoracic wall



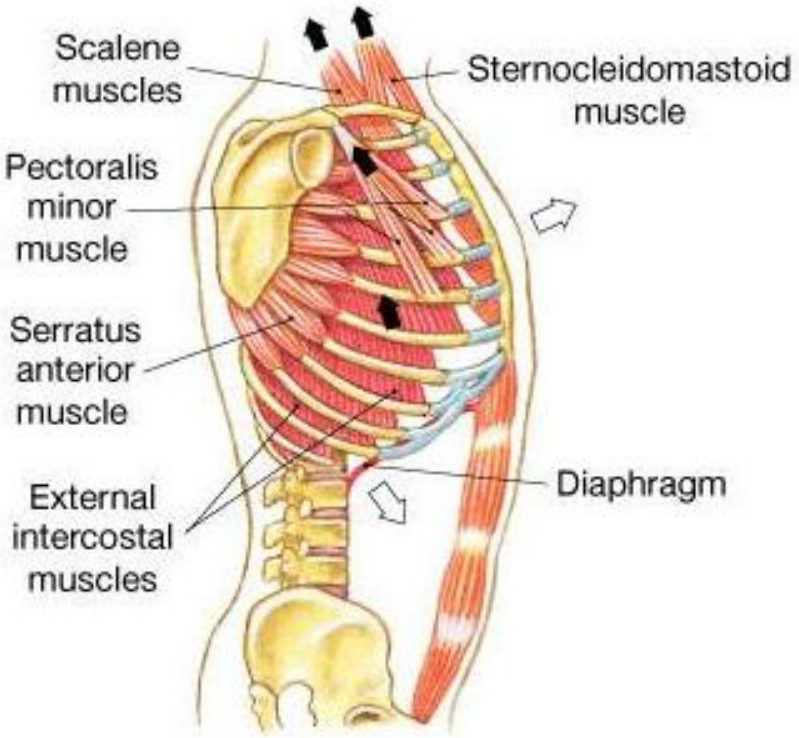
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INSPIRATION

QUIET RESPIRATION

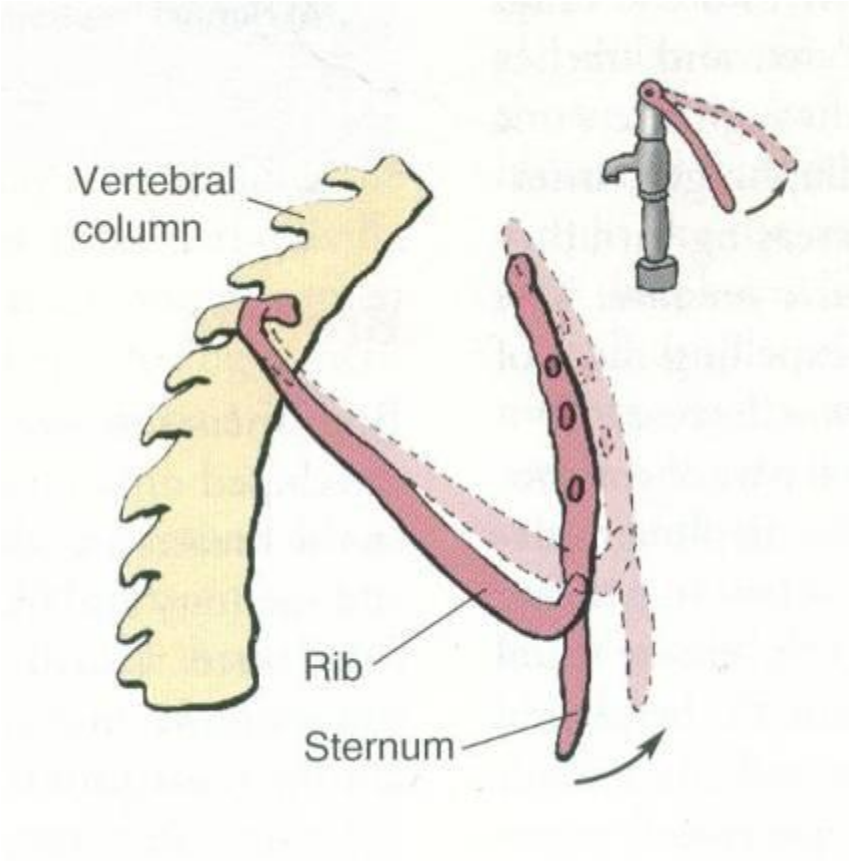
ANTEROPOSTERIOR DIAMETER

- Ribs acting as lever, fulcrum being just lateral to the tubercle
- The anterior end of the rib is lower than the posterior end, therefore, during elevation of the rib, the anterior end also moves forwards
- This occurs mostly in the vertebrosternal ribs



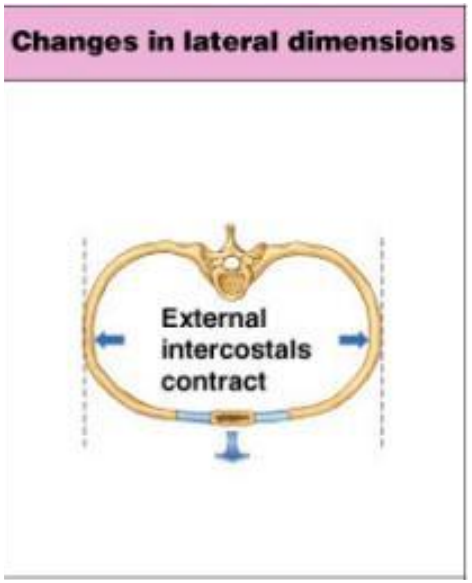
ANTEROPOSTERIOR DIAMETER

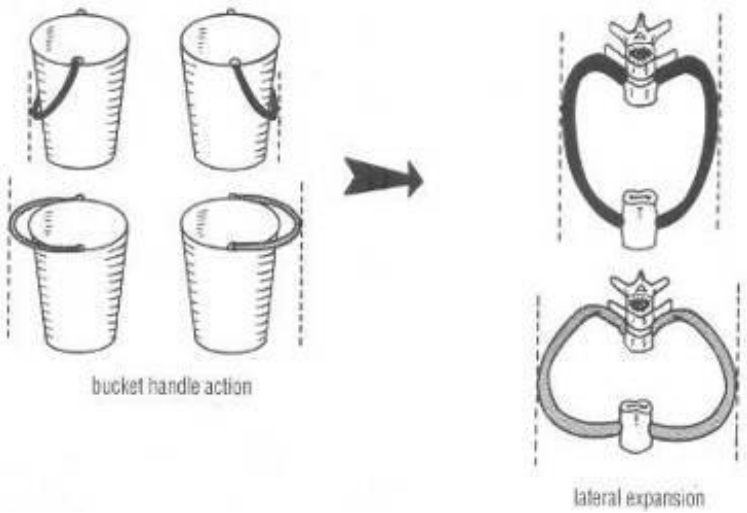
- The body of the sternum also moves up and down
- **'Pump handle movement'.**
- First rib is fixed by contraction of scaleni muscles of the neck and contracting the intercostal muscles
- By this means all the ribs are drawn together and raised toward the first rib



TRANSVERSE DIAMETER

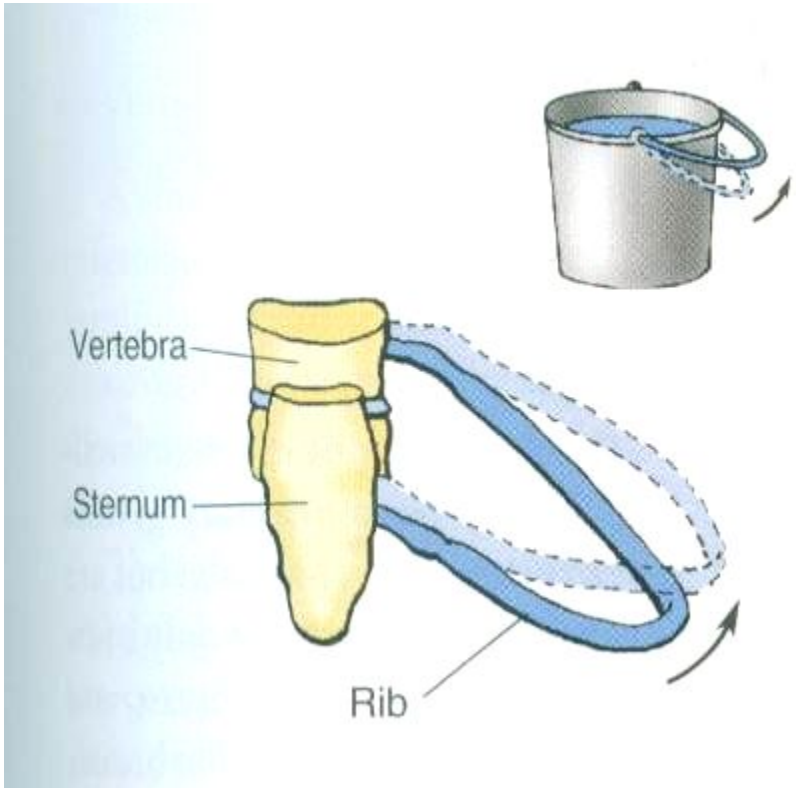
- The ribs curve downwards as well as forwards around the chest wall in this way they resemble **bucket handles**
- During elevation of the rib, the shaft also moves outwards
- If the ribs are raised (like bucket handles),the transverse diameter of the thoracic cavity will be increased





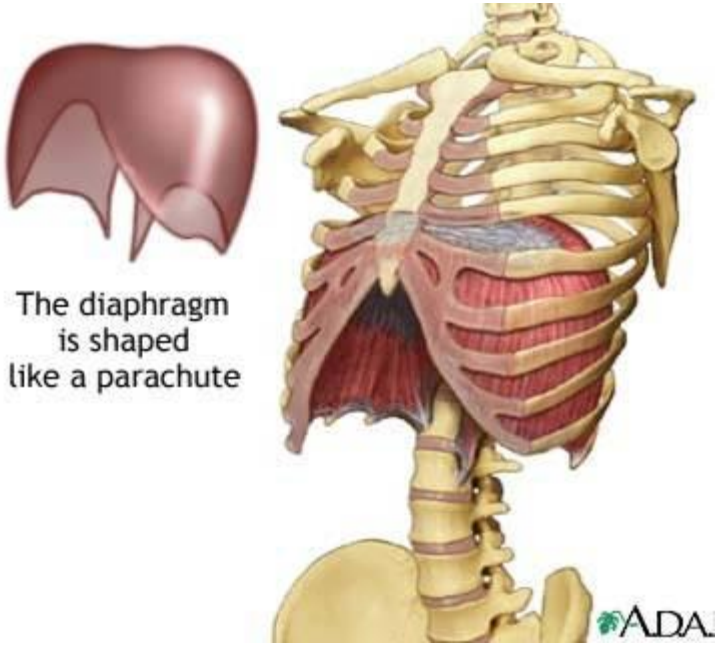
TRANSVERSE DIAMETER

- Transverse diameter is increased by fixing the first rib and raising the other ribs to it by contracting the intercostal muscles
- Mainly in vertebrochondral ribs
- Bucket handle movement



VERTICAL DIAMETER

- To increase vertical diameter there are 2 option
- Either roof is raised or floor is lowered.
- Roof is formed by suprapleural membrane and is fixed
- Floor is formed of mobile diaphragm and when it contracts it becomes flattened and its level is lowered



VERTICAL DIAMETER

- Increased by lowering down of diaphragm
- Slow twitch fibers
- Resistant to fatigue
- Descend in abdomen(1.5-7cm)-increasing vertical diameter of thoracic cavity
- As the diaphragm descends on inspiration,intra-abdominal pressure rises
- This rise in pressure is accommodated by the reciprocal relaxation of the abdominal wall musculature



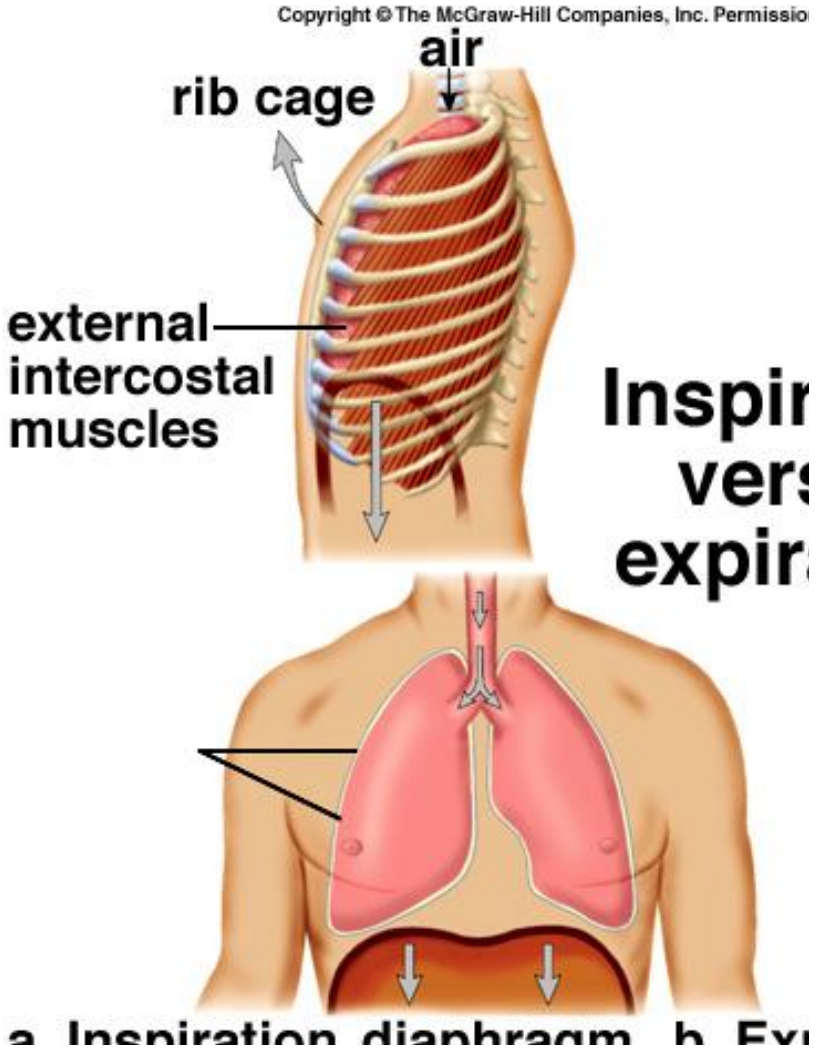
QUIET INSPIRATION

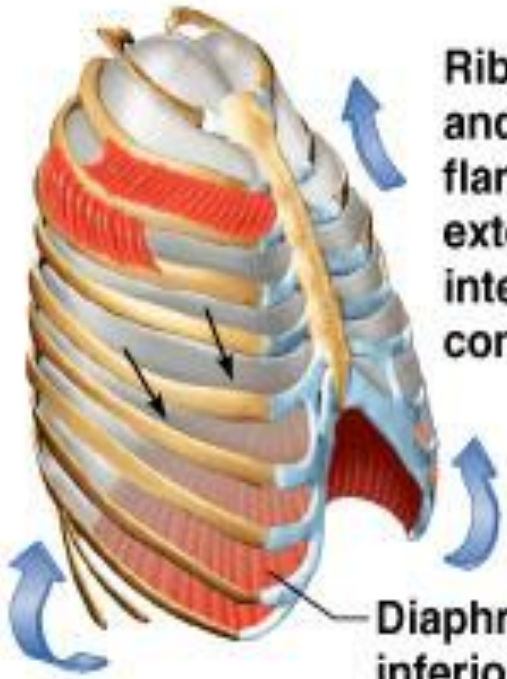
MUSCLES INVOLVED

- Mainly diaphragm
- Intercostal muscles

RESPIRATORY MOVEMENTS

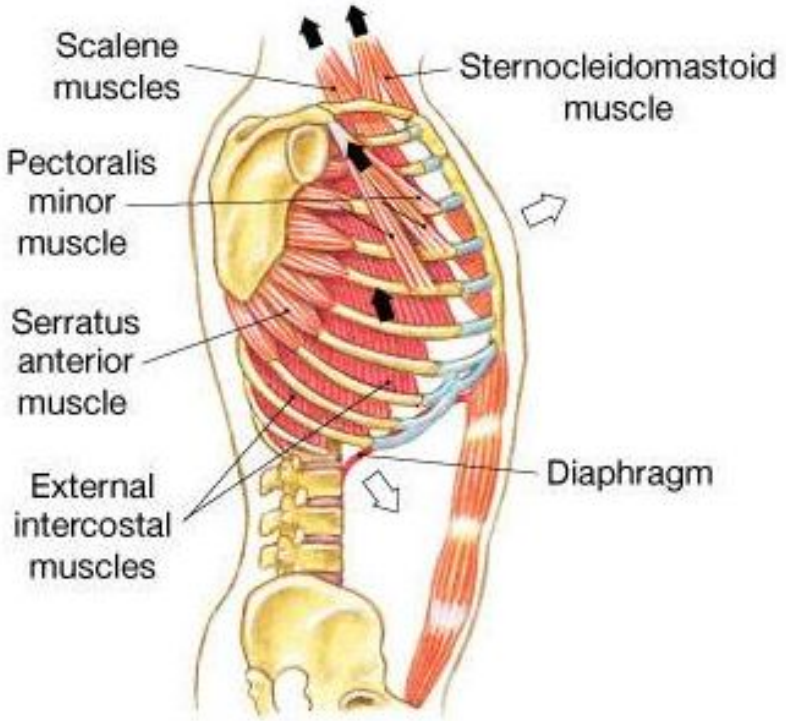
- The anteroposterior diameter of the thorax is increased by elevation of the 2nd to 6th ribs
  - The first rib remains fixed,
  - The transverse diameter is increased by elevation of the 7th to 10th ribs,
  - The vertical diameter is increased by descent of the diaphragm.



	Sequence of events	Changes in anterior-posterior and superior-inferior dimensions
Inspiration	① Inspiratory muscles contract (diaphragm descends; rib cage rises) ↓ ② Thoracic cavity volume increases ↓ ③ Lungs stretched; intrapulmonary volume increases ↓ ④ Intrapulmonary pressure drops (to –1 mm Hg) ↓ ⑤ Air (gases) flows into lungs down its pressure gradient until intrapulmonary pressure is 0 (equal to atmospheric pressure)	 Ribs elevated and sternum flares as external intercostals contract Diaphragm moves inferiorly during contraction

DEEP INSPIRATION

- Movements during quiet inspiration are increased
- The first rib is elevated directly by the scaleni, and indirectly by the sternomastoids
- The concavity of the thoracic spine is reduced by the erector spinae.

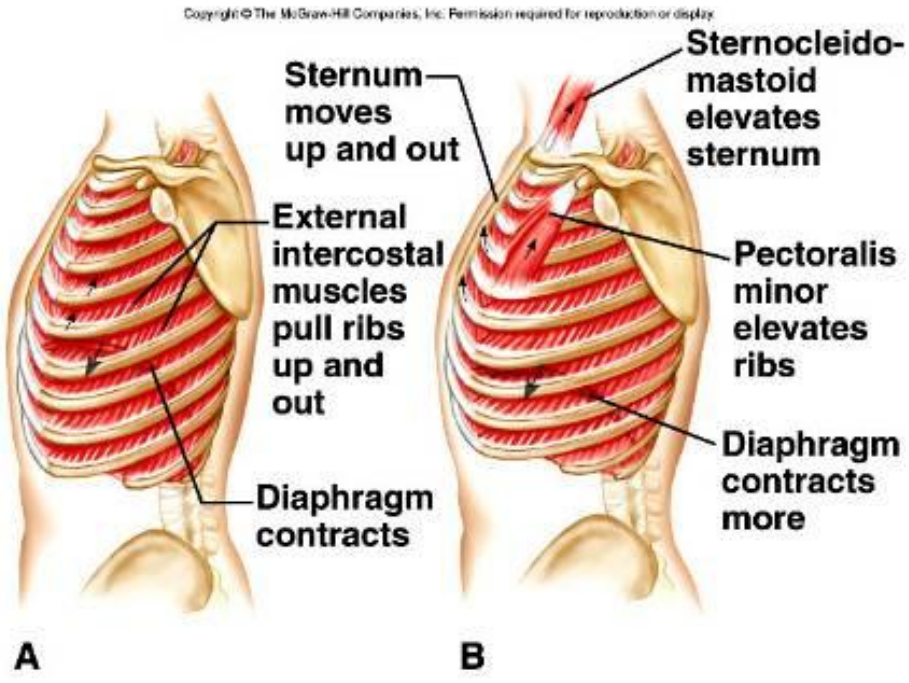


MUSCLES INVOLVED IN FORCED INSPIRATION

- Diaphragm
- The intercostal muscles
- The sterno-mastoids
- The scaleni
- The serratus anterior, the pectoralis minor, and the erector spinae
- The alaequae nasi open up the external nares.

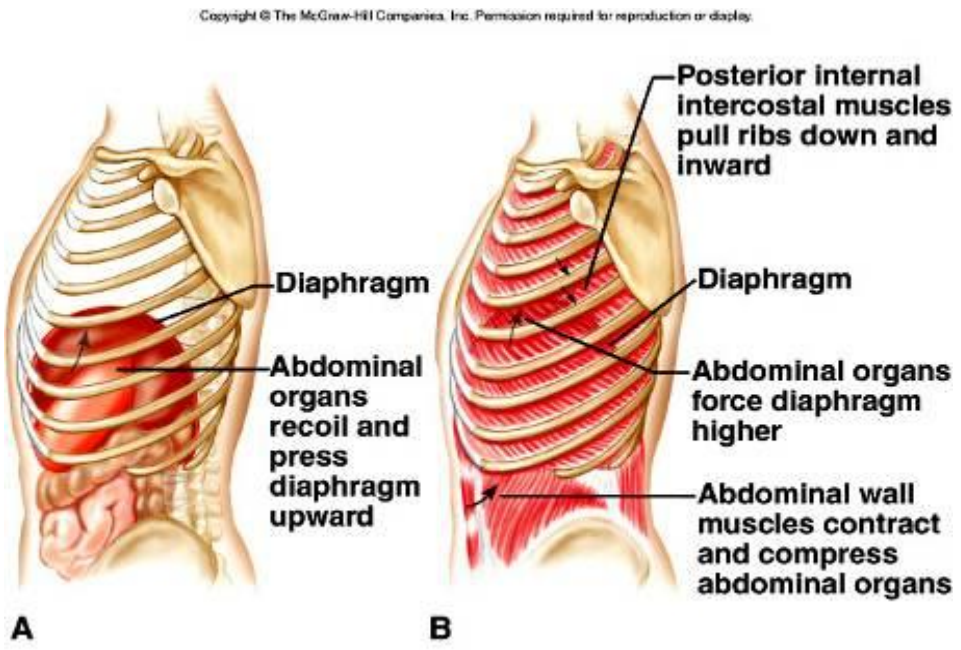
RESPIRATORY MOVEMENTS IN FORCED INSPIRATION

- A maximum increase in the capacity of the thoracic cavity occurs
- Every muscle that can raise the ribs is brought into action including the scalenus anterior and medius and sternocleidomastoid serratus anterior and pectoralis major



QUIET EXPIRATION

- A passive process
- Elastic recoil of the lungs
- Relaxation of diaphragm and external intercostals muscles
- **Diaphragm** moves upwards
- Increase in tone of muscles of anterior abdominal wall
- Decrease in all dimensions of chest



Expiration

① Inspiratory muscles relax (diaphragm rises; rib cage descends due to recoil of costal cartilages)

↓

② Thoracic cavity volume decreases

↓

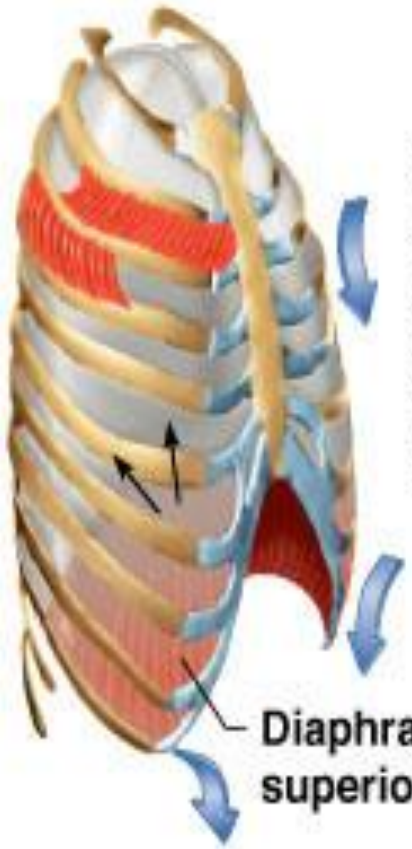
③ Elastic lungs recoil passively; intrapulmonary volume decreases

↓

④ Intrapulmonary pressure rises (to +1 mm Hg)

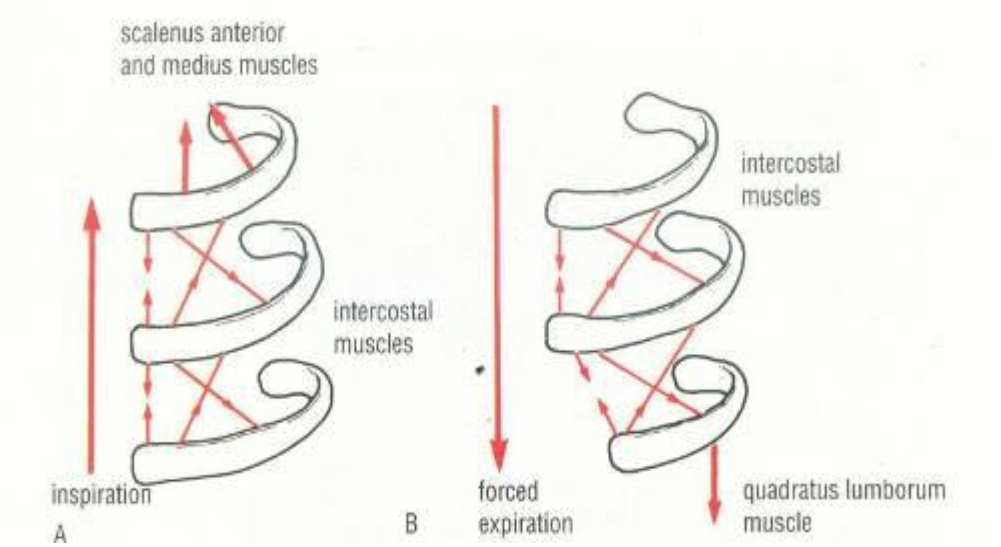
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⑤ Air (gases) flows out of lungs down its pressure gradient until intrapulmonary pressure is 0

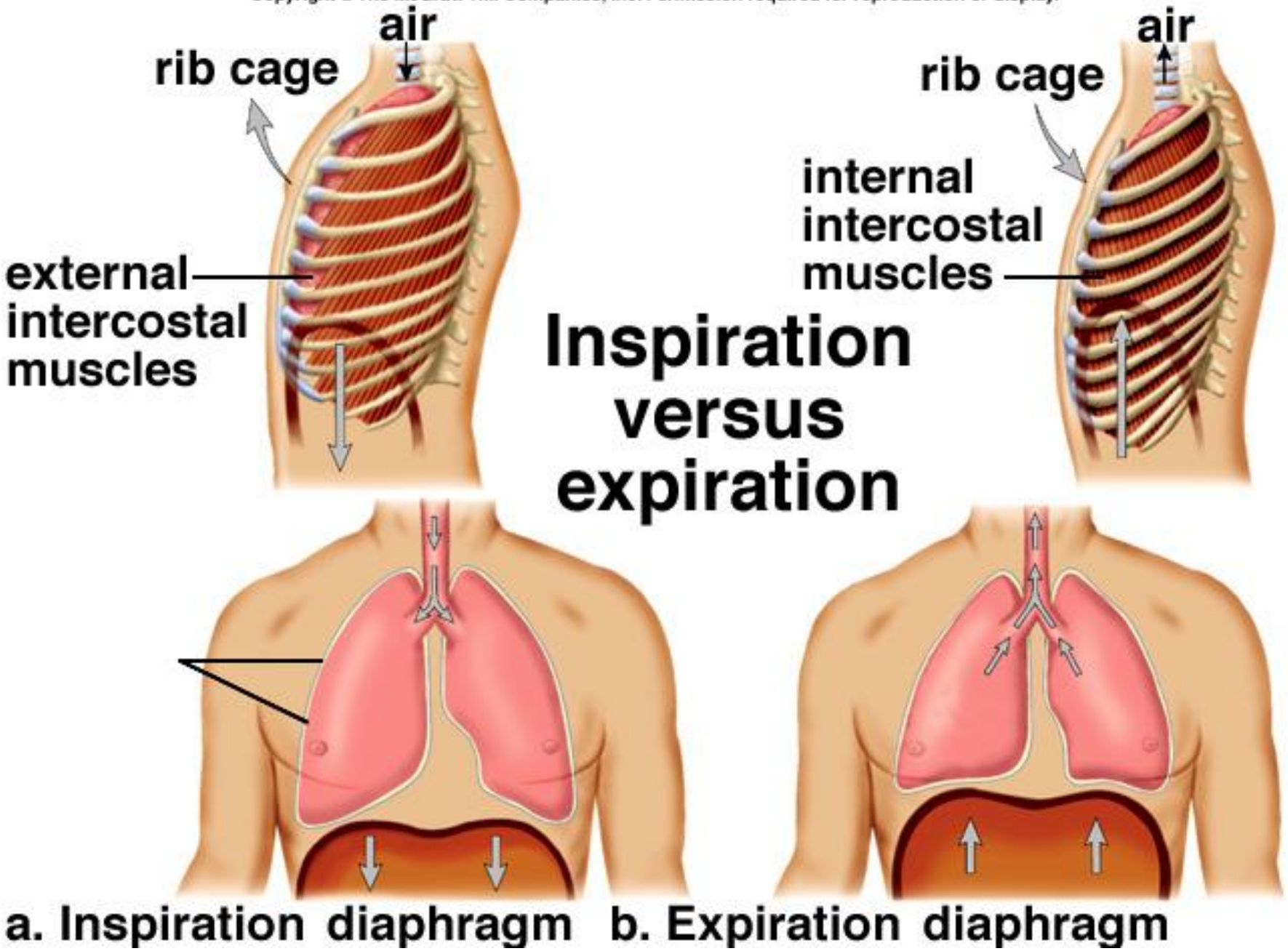


FORCED EXPIRATION

- An active process
- By forcible contraction of muscles of anterior abdominal wall
- **Quadratus lumborum** contracts and pulls down the twelfth ribs
- **Intercostal muscles** pulls the ribs together and depress them to the lowered twelth rib
- **Serratus posterior inferior and latissimus dorsi are also involved**



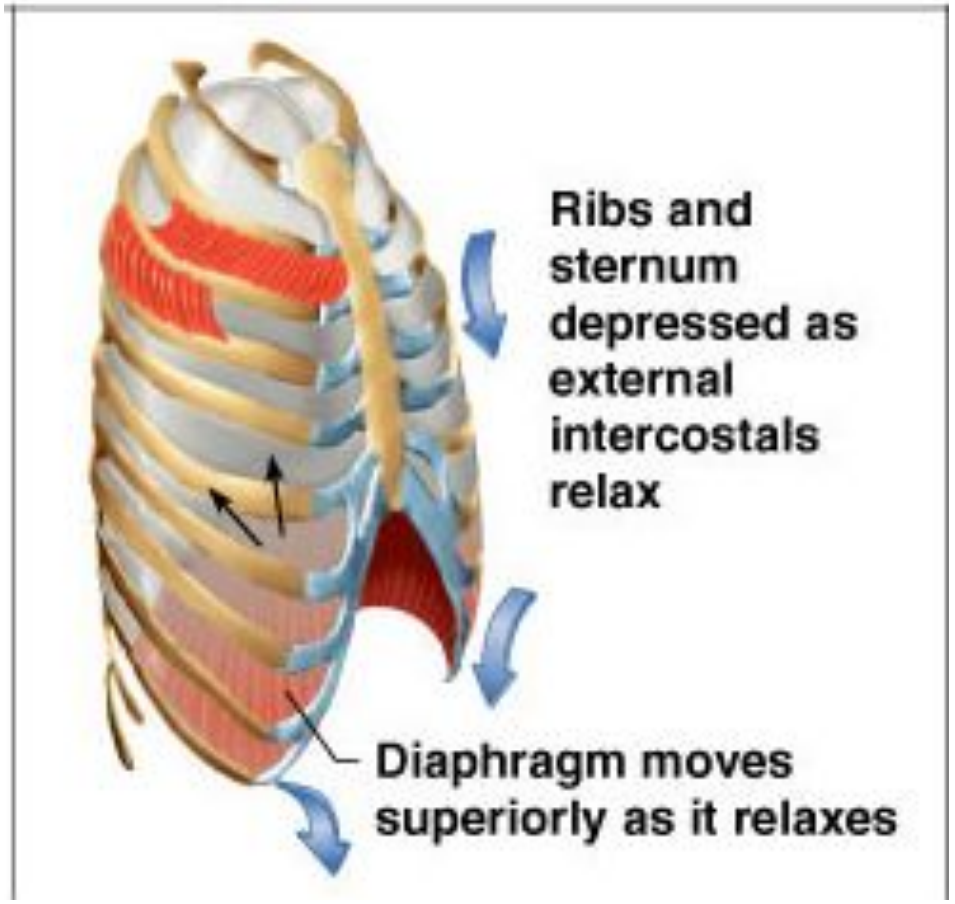
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CLINICAL CORRELATES

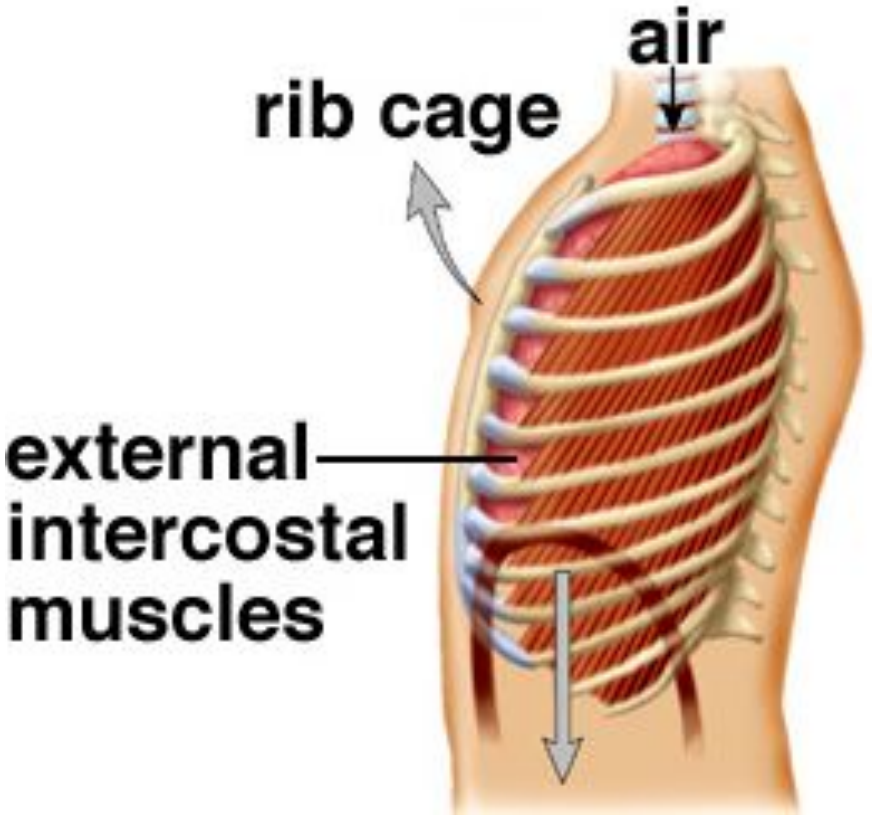
- In dyspnea (breathlessness, difficult breathing) the patients are most comfortable on sitting up, leaning forwards and fixing the arms
- In the sitting posture the position of diaphragm is lowest allowing maximum ventilation
- Fixation of the arms fixes the scapulae, so that the serratus anterior and pectoralis minor may act on the ribs to good advantage.
- The height of the diaphragm in the thorax is variable according to the position of the body and tone of the abdominal muscles
- It is highest on lying supine, high on standing, and lowest on sitting down

Which phase of respiration is this??



EXPIRATION

Which phase of respiration??



THANK YOU